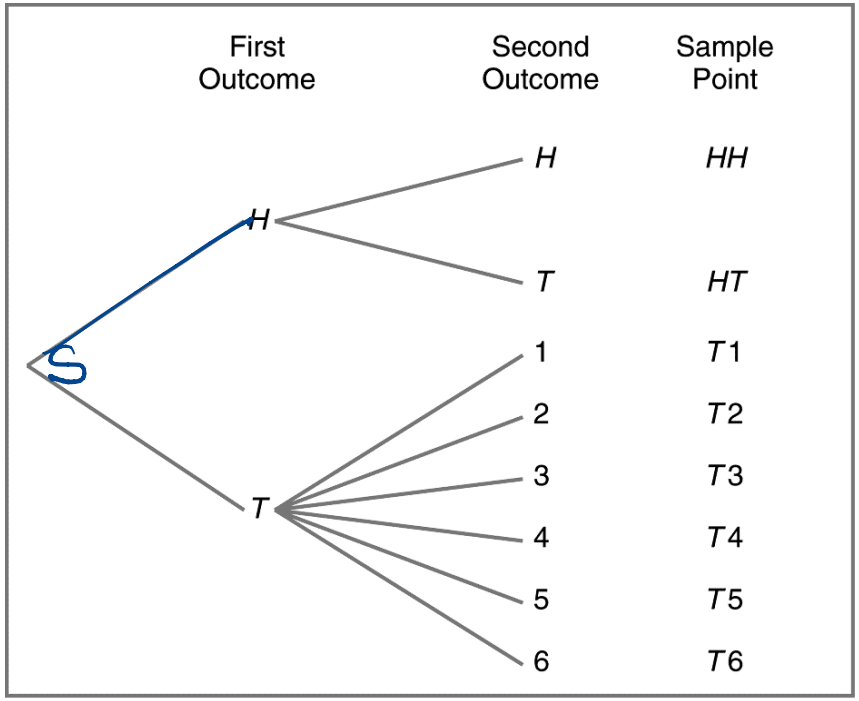
Tree diagram

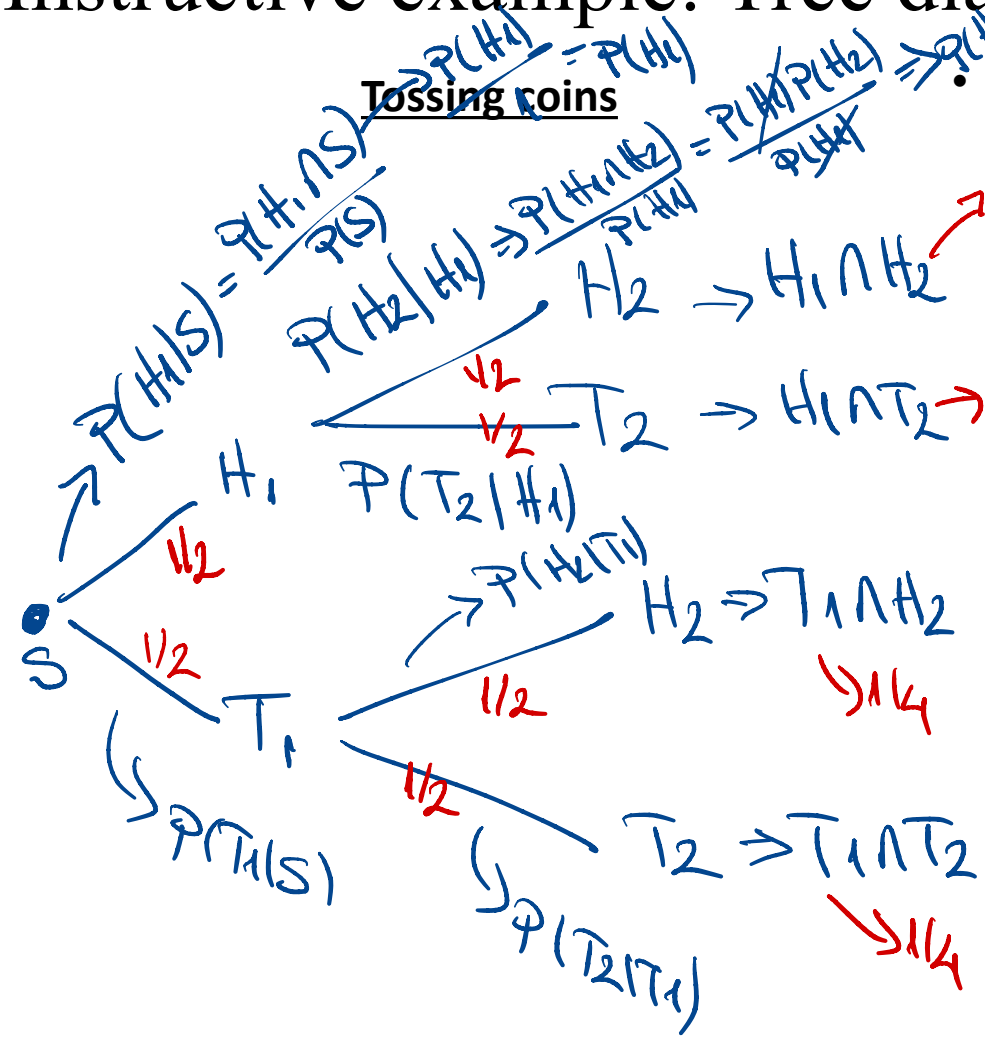
- List all the possible outcomes systematically by means of tree diagram

An experiment consists of flipping a coin and then flipping it a second time if a head occurs. If a tail occurs on the first flip, then a die is tossed once



Instructive example: Tree diagram

Tossing coins



- **Experiment: Tossing fair coins**
- **Multiply** the probabilities along **branches**
- The total probability for all outgoing branches from one node is 1
- The probability of getting two Head from two tosses 1/4
- The probability of getting one Head and one Tail from two tosses 1/4 + 1/4 = 1/2
- The probability of getting at least one Head from two tosses 3/4

Example 1.7

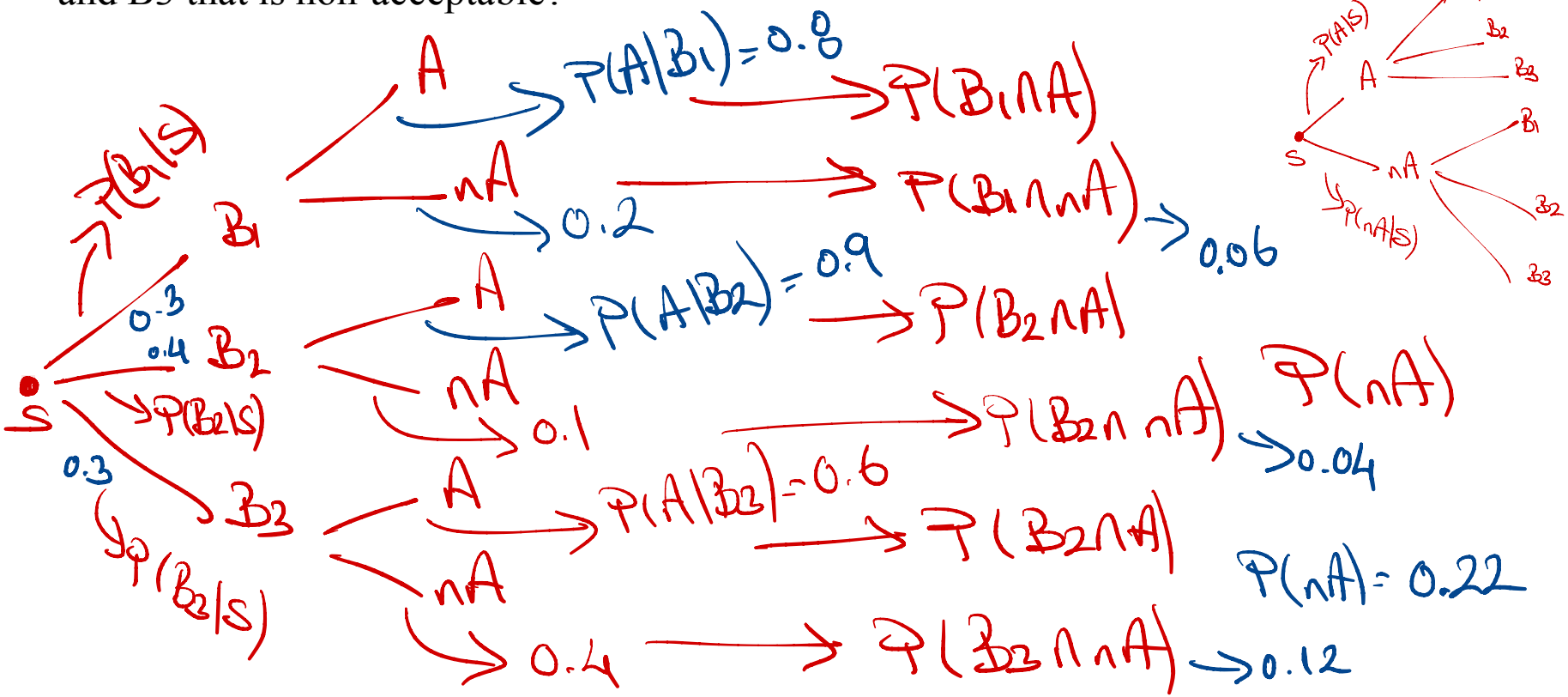
For the resistor in Example 1.5, we used "A" to denote the event that a randomly chosen resistor is acceptable. Now, we use the notation NA (Non-acceptable) for the complement of A. What is the probability of choosing a resistor from machine B1, B2 and B3 that is non-acceptable?

Example 1.5

A company has three machines B_1, B_2, and B_3 making 1 kΩ resistors. Resistors within 50 Ω of the nominal value are considered acceptable. It has been observed that 80% of the resistors produced by B_1 and 90% of the resistors produced by B_2 are acceptable. The percentage for machine B_3 is 60%. Each hour, machine B_1 produces 3000 resistors, B_2 produces 4000 resistors, and B_3 produces 3000 resistors. All of the resistors are mixed together at random in one bin and packed for shipment. What is the probability that the company ships an acceptable resistor?

Example 1.7

For the resistor in Example 1.5, we used "A" to denote the event that a randomly chosen resistor is acceptable. Now, we use the notation NA (Non-acceptable) for the complement of A. What is the probability of choosing a resistor from machine B1, B2 and B3 that is non-acceptable?



Counting Methods

If the sample space S has finite # of equally likely outcomes

$$P(A) = \frac{\#\ of\ elements\ in\ A}{\#\ of\ elements\ in\ S}$$

Counting Methods

Fundamental principles of counting methods;

- Multiplication Rule
- Combination Rule
- Permutation Rule

Multiplication rule

- Let A be an experiment and $A_1, A_2, \ldots A_k$ be subexperiments of A with $\#n_1$, $\#n_2 \ldots, \#n_k$ outcomes, respectively.

The experiment A has $\prod_{i=1}^{k} n_i$ outcomes.

$2 \times 3 = \underline{\underline{6}}$

Instructional Example

3 2 1

- Consider three letters

a, b, c

How many different arrangements are possible?

The number of permutations of n objects is n!

$\underline{4}\ \underline{3} = \underline{\underline{12}}$

- Consider four letters

a, b, c, d

$ab \neq ba$

4 letters taken 2

$$ {}_nP_k = \frac{n!}{(n-k)!} $$

Example 1.8

How many possible license plates could be stamped if each license plate were required to have exactly 3 letters and 4 numbers?

26 letters

A	A	A	0	0	0	0
B	B	B	1	1	1	1
C	C	C	2	2	2	2
⋮	⋮	⋮	⋮	⋮	⋮	⋮
			9	9	9	9
Z	Z	Z				

10 numbers

26	26	26	10	10	10	10

Example 1.9

How many possible license plates could be stamped if each **license plate were required to have 3 unique letters and 4 unique numbers**?

A			0			
B	B		1	1		
C	C	C	2	2	2	3
⋮	⋮	⋮	⋮	⋮	⋮	⋮
			9	9	9	9
Z	Z	Z				

26 25 24 10 9 8 7

Example 1.10

Choose 7 cards at random from a deck of 52 different cards. Display the cards in the order in which you choose them. How many different sequences of cards are possible?

$\underline{52}\ \underline{51}\ \underline{50}\ \underline{49}\ \underline{48}\ \underline{47}\ \underline{46}$

P^{52}_{7}

Example 1.11

Suppose there are four objects, A, B, C, and D. Consider an experiment in which the procedure is to choose two objects without replacement. How many possible outcomes do you have?

4 3

Example 1.12

Suppose there are four objects, A,B, C and D. Define an experiment in which the procedure is to choose two objects without replacement and **arrange them in alphabetical order**. How many possible outcomes of the experiment ?

$$\frac{\underline{4}\ \underline{3}}{2!} = 6 \Rightarrow C^4_2$$

$$\binom{n}{k} = \frac{n!}{k!\ (n-k)!}$$

Example 1.13

There are 52 cards in a deck. You are given seven cards at random from the deck. After receiving each card **you return it to the deck and receive another card at random**. How many possible outcomes of the experiment ?

$$\underline{52}\ \underline{52}\ \underline{52}\ \underline{52}\ \underline{52}\ \underline{52}\ \underline{52}$$

$$= \underline{\underline{52^7}}$$

Example 1.14

There are four queens in a deck of 52 cards. You are given seven cards at random from the deck. After receiving each card you return it to the deck and receive another card at random. Observe whether you have not received any queens among the seven cards. How many possible outcomes of the experiment ?

What is the probability that you have not received any queens ?

$$48^7 / 52^7$$

A bin of 50 manufactured parts contains 3 defective parts and 47 nondefective parts. A sample of 6 parts is selected without replacement. That is, each part can be selected only once, and the sample is a subset of the 50 parts. How many different samples are there of size 6 that contain exactly 2 defective parts?

3 2 47 46 49 44 ⇒ permutation

but the order is not important

$$C^{3}_{2}\, C^{47}_{4}$$

Detailed Solution